# IN THE NOETHERIAN DESCENT SITUATION, THE RESOLUTION PROPERTY IS EQUIVALENT TO THE PUSHFORWARD IDEAL BEING A FINITE LOCALLY FREE QUOTIENT

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 13537-13543 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 13545-13563, the complete author proof (17 body lines).

*Editorial note.* Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Noetherian reduction step of the spaces resolution-property descent series (feeding the blow-up capstone family L16104/L16134).

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. Cohomology of Spaces, lemma finite-pushforward-coherent; the chapter's Situation setup (situation-descend-resolution-property) and its preceding lemma surjection-in-Noetherian-case (the generation criterion from the powers of I).

*Source reference.* Original lemma 0GUZ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13440–13450. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0GUY: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13416–13437. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 07UH: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13563–13563. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *In Situation 0GUY, assume $X$ is Noetherian. Then $X$ has the resolution property if and only if $\pi_*\mathcal{I}$ is the quotient of a finite locally free $\mathcal{O}_X$-module.*

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*Date*: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Proof.* The module $\pi_*\mathcal{I}$ is coherent by Cohomology of Spaces, Lemma 07UH. Hence if $X$ has the resolution property then $\pi_*\mathcal{I}$ is the quotient of a finite locally free $\mathcal{O}_X$-module. Conversely, assume given a surjection $\mathcal{E} \to \pi_*\mathcal{I}$ for some finite locally free $\mathcal{O}_X$-module $\mathcal{E}$. Observe that for all $n \geq 1$ there is a surjection

$$\pi_*\mathcal{I} \otimes_{\mathcal{O}_X} \pi_*\mathcal{I}^n \longrightarrow \pi_*\mathcal{I}^{n+1}$$

Hence $\mathcal{E}^{\otimes n}$ surjects onto $\pi_*\mathcal{I}^n$ for all $n \geq 1$. We conclude that $X$ has the resolution property if we combine this with the result of Lemma 0GUZ. $\square$